

UNDERSTANDING

SOFTWARE DEVELOPMENT
LIFE CYCLE

TESTING AUTOMATION TOOLS

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What is SDLC?

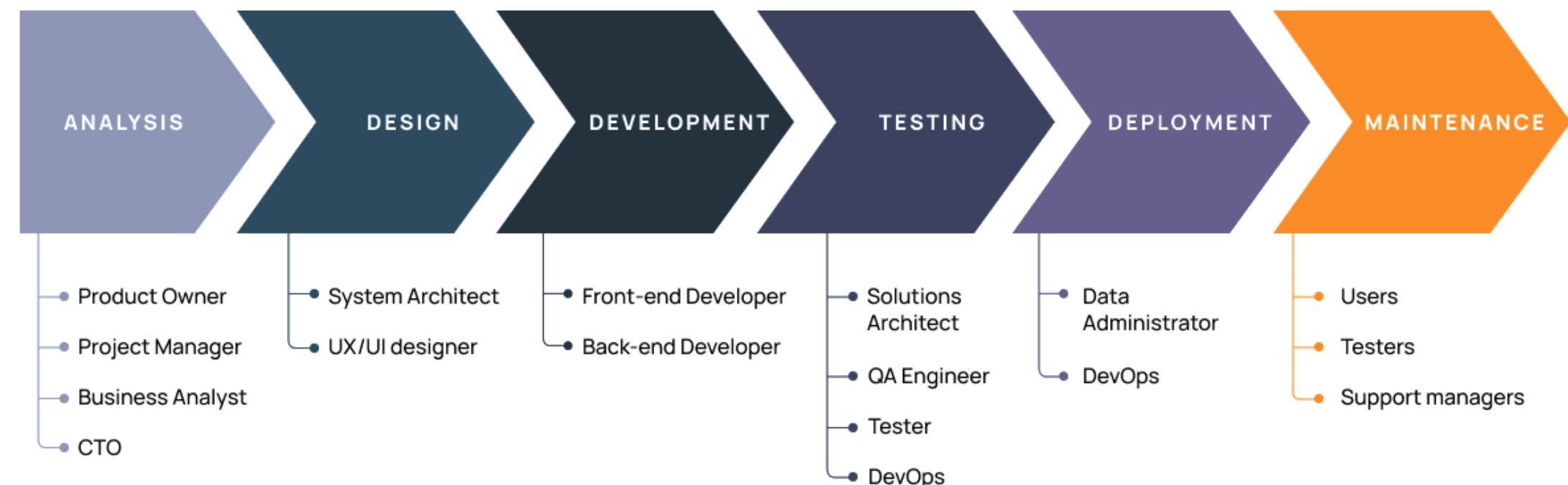


SDLC (Software Development Life Cycle) is a systematic, step-by-step process used to develop software that meets technical, business, and user expectations.

Purpose:

- Deliver high-quality software
- Meet project deadlines and budget
- Reduce risks and errors
- Provide a clear roadmap for teams

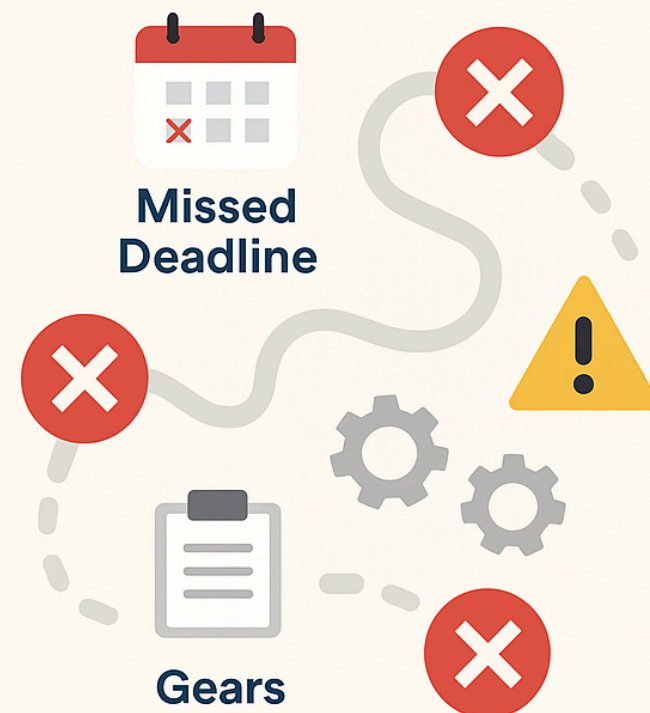
6 Phases of the Software Development Life Cycle



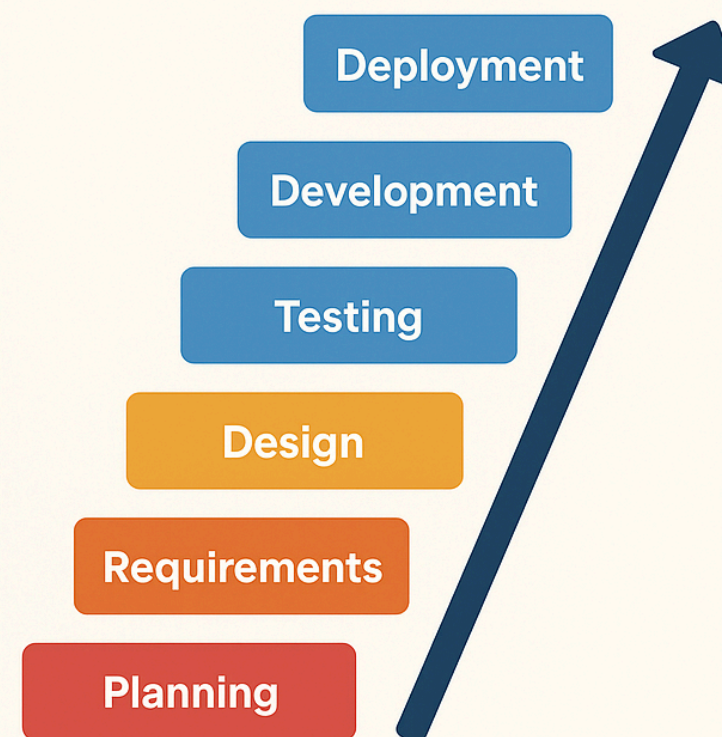
- Organizes the development process – eliminates guesswork and random coding
- Aligns stakeholders – ensures clients, developers, testers, and managers are on the same page
- Improves risk management – identifies potential problems early
- Provides a roadmap – avoids chaos by guiding the team through structured phases

Why SDLC Matters

CHAOTIC PROCESS



STRUCTURED SDLC



Without SDLC, teams often face scope creep, miscommunication, and delays. SDLC acts like a blueprint, ensuring structure and predictability.

List of Phases with Brief Descriptions:

- **Planning** – Define goals, scope, timeline, and team roles
- **Requirements** – Gather and document user needs and system expectations
- **Design** – Create architecture, user interfaces, and system models
- **Development** – Write and assemble the actual code
- **Testing** – Detect and fix bugs; validate functionality
- **Deployment** – Release the software for use
- **Maintenance** – Monitor performance, apply updates, and fix issues over time

Benefits of Using SDLC

Higher Quality Output

Structured steps ensure fewer bugs and better performance



Predictable Timelines & Costs

Better planning leads to fewer delays and cost overruns



Clear Roles & Deliverables

Each team member knows what to do, when, and how



Improved Customer Satisfaction

Software meets real user needs and works reliably



✓ **Better Quality**

Fewer bugs, smoother performance

🕒 **Predictable Delivery**

Clear timelines and controlled costs

👤 **Defined Roles**

Everyone knows their responsibilities

★ **Happy Customers**

Software meets real user needs

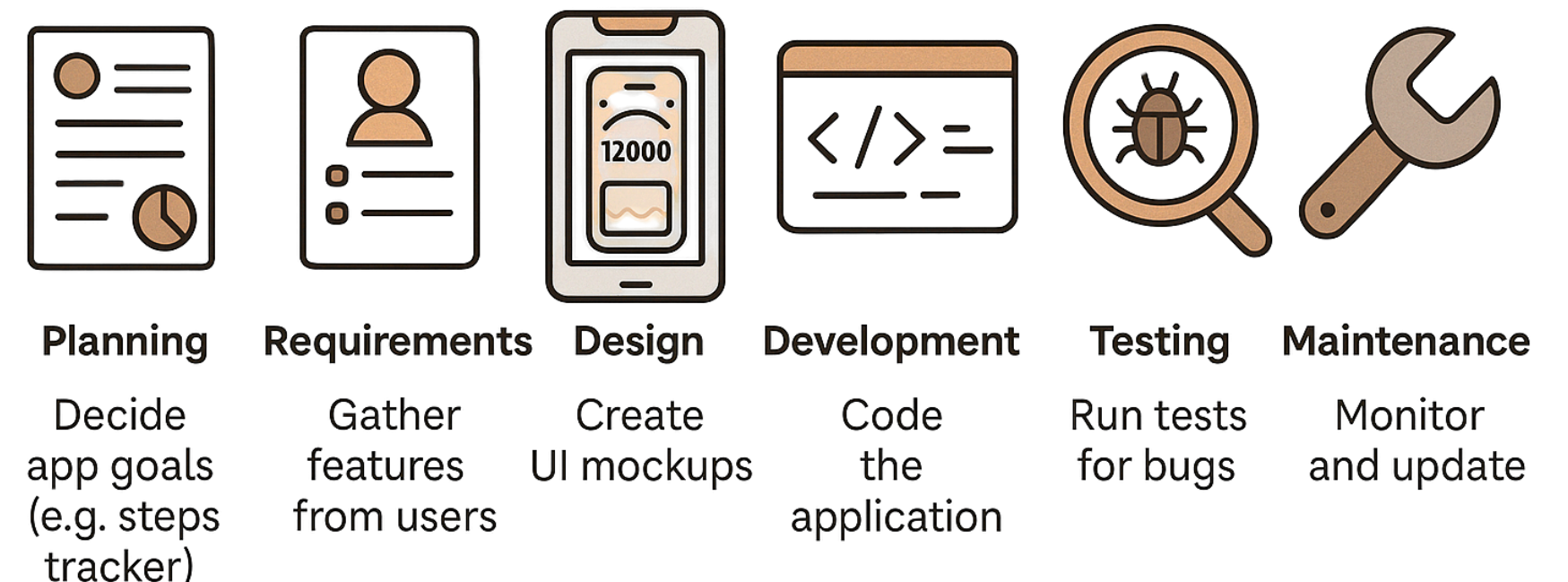
Scenario: Building a Fitness Tracker App

🧩 Phase-wise Breakdown:

- Planning: Decide app goals – e.g., steps tracker, water intake, reminders
- Requirements: Gather features from users – login, daily goals, sync with wearable
- Design: Create wireframes and UI mockups using Figma
- Development: Code frontend (React Native) and backend (Node.js, MongoDB)
- Testing: Run unit tests and real-device testing for bugs
- Deployment: Release on App Store / Google Play
- Maintenance: Monitor crash reports, push updates, fix bugs

SDLC in Action: A Mobile App Example

Fitness Tracker App



Amazon Web Services (AWS)

“What is SDLC?”

<https://aws.amazon.com/what-is/sdlc>

Atlassian Agile Coach

“Software Development Life Cycle (SDLC)”

<https://www.atlassian.com/agile/software-development/sdlc>

Harness Blog

“Software Development Life Cycle Phases”

<https://www.harness.io/blog/software-development-life-cycle-phases>

“SDLC isn’t just a process — it’s the backbone of successful software.”

**Thank
you**